

ASSIGNMENT

Due Date: Week 13 - Monday 12th-October-2026, 10:00am Perth time

Weight: 40% of the unit mark

Use of AI: You may use AI tools to support learning and understanding programming concepts. However, vibe coding is not permitted. In this assignment, vibe coding means submitting code that you do not understand or cannot explain, including code generated primarily by an AI tool with little or no meaningful student design, verification, or documentation. If an AI tool assisted with any part of your submitted code, you must include clear comments in the relevant code cells or source-code sections indicating which parts were AI-assisted and how the tool was used.

1. Overview

In this assignment, you will solve a real-world data mining problem. This assignment requires you to understand the theory discussed in the workshops, conduct some research into the data mining problem to solve, and use the skills that you should have developed through completing practical exercises to perform various data mining tasks.

Please note that this is an individual assignment. Whilst you may discuss general data mining topics related to this assignment with other students, you must make sure that your work is not accessible by anyone else. There are many choices to make and therefore it is very unlikely to have identical submissions by chance. Submissions that are very similar will be investigated for academic misconduct.

2. Problem Description

In this assignment, you will undertake a predictive task involving the classification of network activities. A set of four files has been provided:

- **Training-data.xlsx** – labelled dataset for model development and training.
- **Test-data-1.xlsx** and **Test-data-2.xlsx** – datasets for final evaluation of the selected classifier.
- **Feature-description.xlsx** - descriptions of the attributes contained in the released datasets.

The training dataset comprises records representing both *normal activities* and *malicious activities*. The core objective is to build a binary classification model that predicts the target variable label: Normal activity and Malicious/attack activity.

The dataset contains both categorical and numerical attributes and may require preprocessing. You should investigate issues such as missing or placeholder values, duplicates, irrelevant or redundant attributes, inappropriate data types, scaling requirements, and class imbalance. Any preprocessing decisions must be clearly justified. The assignment consists of three main components:

1. **Data Preparation:** careful examination and preprocessing of the dataset to resolve quality issues. This includes handling missing values, duplicates, irrelevant or redundant attributes,

inappropriate data types, normalisation/standardisation of numerical features, class imbalance, and potential feature engineering. The outcome of this phase is a clean, well-structured dataset ready for model development.

2. **Data Classification:** development of suitable classification models using the training dataset. This requires analytical processing, selection and tuning of classifiers, and rigorous validation (e.g., cross-validation). Models must then be evaluated and compared, with the selected final model applied to classify the two test datasets.
3. **Report:** a comprehensive report must document the entire process, from data preparation to classification. Each decision must be clearly explained and justified (e.g., imputation strategies for missing values, reasons for classifier selection, handling of class imbalance). The report should include evaluation metrics such as confusion matrices, accuracy, precision, recall, and F1 score.

3. The Tasks

3.1 Data Preparation

The first stage of the assignment involves a systematic examination of the dataset in order to identify and address potential data quality issues. The outcome of this process is a cleaned and pre-processed training dataset, which will serve as the foundation for subsequent classification tasks. Marks will be awarded based on the depth and rigor with which issues are identified, addressed, and justified. The key aspects to consider include:

Irrelevant Attributes

- Investigate whether any attributes are irrelevant or redundant for the prediction task.
- Describe what you think irrelevant attributes are.
- For each attribute, carefully examine it and decide whether it is irrelevant. If so, give a brief explanation and remove the attribute.

Missing Values

- Which attributes/instances have missing entries?
- For those attributes/instances, how many missing entries are present?
- For each attribute/instance with missing entries, make a suitable decision, justify it, and proceed.

Duplicates

- Detect if there are any duplicates (instances/attributes) in the original data?
- For each attribute/instance with duplicates, make a suitable decision, justify it, and proceed.

Data Types

- Review the data types (e.g., numerical, categorical) assigned to each attribute.
- If the data type of an attribute is not suitable, give a brief explanation and convert the attribute to a more suitable data type. Provide detailed information of the conversion.

Scaling and Standardisation

- For each numeric attribute, decide if any pre-processing (e.g. scaling, standardisation) is required. Explain why it is needed (this should be discussed in relation to the subsequent classification task).

Feature/Attribute Selection

- if applicable, clearly indicate which attributes you decide to remove in addition to those (obviously) irrelevant attributes that you have identified above and give a brief explanation why.

Data Instances

- if you decide to make changes to the data instances with class labels (this may include selecting only a subset of the data, removing instances, randomizing/reordering instances, or synthetically injecting new data instances to the training data, etc.), provide an explanation.

Class Imbalance

- Examine class imbalance and its potential impact. If corrective action is taken, justify the approach; otherwise, justify why no action is necessary.

Feature Engineering

- you may also come up with attributes derived from existing attributes. If this is the case, explain the new attributes that you have created.

Other Preprocessing Steps

- Describe any additional preprocessing actions not covered above.

3.2 Data Classification

This task requires the design, evaluation, and comparison of classification models. The overall objective is to select appropriate classifiers, validate their performance, and predict whether each network activity is Normal or Malicious/Attack.

Classifier Selection

- At least three classifiers must be implemented: **k-Nearest Neighbour (k-NN)**, **Naïve Bayes**, and **Decision Trees**.
- Additional classifiers, including ensemble or meta-classifiers, may also be considered.
- For each classifier, clearly document any parameter tuning undertaken to improve performance (e.g., number of neighbours in k-NN, pruning for decision trees, or smoothing in Naïve Bayes). Explain the meaning of each parameter and justify the chosen values.

Cross-Validation

Although test datasets are provided, cross-validation on the training data is required. The following must be addressed:

- Justify your choice of validation/cross-validation strategy.
- Use cross-validation to evaluate and compare classifiers, tune hyperparameters, and assess preprocessing, feature-selection, or class-imbalance strategies where applicable.
- For each classifier, report and interpret the cross-validation performance using appropriate evaluation metrics and confusion matrices.
- Any learned preprocessing step, such as imputation, scaling, feature selection, or resampling, must be fitted within each training fold to avoid data leakage.

Classifier Comparison

- Compare the classification performance of the candidate classifiers using at least two appropriate evaluation metrics, such as accuracy and F1-score.
- Based on the cross-validation results, identify the best-performing classifiers and justify the selection of one final model.
- Clearly state the final preprocessing pipeline and parameter settings used by the selected model.

Test dataset Evaluation

- After the final classifier, preprocessing pipeline, hyperparameters, and any feature-selection strategy or decision threshold where applicable have been fixed, train the final model using all available labelled instances in Training-data.xlsx.
- Apply the final model to Test-data-1.xlsx and Test-data-2.xlsx, and report the classification performance on both test datasets.
- The test datasets must be used for final evaluation only and must not be used for classifier selection, hyperparameter tuning, threshold selection, feature selection, or preprocessing decisions.

The label column in the test datasets is provided only for final performance evaluation and report writing. It must not be used as an input predictor, nor used for training, model selection, hyperparameter tuning, threshold selection, feature selection, or preprocessing decisions.

Generate two prediction files: Predict_1.xlsx and Predict_2.xlsx, corresponding to the two test datasets. Because the test datasets do not contain a separate ID column, the ID in each prediction file must be the row index of the test instance, starting from 1 for the first data row after the header. The predict column must use the same coding as the target variable: 0 = Normal and 1 = Malicious/Attack. Each file must be strictly formatted as follows:

ID	predict
1	0
2	1
3	1
...	...

Predict_1.xlsx

ID	predict
1	0
2	0
3	1
...	...

Predict_2.xlsx

IMPORTANT: please ensure that your prediction is correctly formatted as required. Predict_1.xlsx must contain 59,807 predictions with IDs 1 to 59,807, and Predict_2.xlsx must contain 58,021 predictions with IDs 1 to 58,021. Your marks will be deducted if your prediction file does not meet the above requirements. No correction to the prediction is allowed after your assignment is submitted.

3.3 Report

You are required to submit a written report that fulfills the following objectives:

- Demonstrates your understanding of the problem and the steps you undertook to solve the tasks.
- Provides sufficient information for the assessment of your work.

Report Restrictions

- **Page limit:** The report must not exceed **15 pages (excluding the reference)**. Any content beyond page 15 will be disregarded during marking.

Structure of the report

- **Cover Page**
 - Include your full name, student ID, and unit details.

- On the cover page of your report, clearly state the average classification accuracy achieved by your single final selected model across Test-data-1.xlsx and Test-data-2.xlsx.
- **Summary**
 - Provide a brief overview of the major findings from your work (e.g., data preparation steps, classification outcomes).
 - Reflect on lessons learned throughout the assignment.
- **Methodology:** address the requirements described above for
 - Data Preparation.
 - Data Classification.
- **Prediction Results**
 - Present a table comparing the cross-validation performance of your two best-performing classifiers.
 - Clearly identify and justify the single final model selected.
 - Report the final selected model's performance on Test-data-1.xlsx and Test-data-2.xlsx.
- **References**
 - Include any references to textbooks, academic papers, online documentation, or tools you consulted during the assignment.

Visual Support

- Use tables, figures, plots, diagrams, or screenshots to illustrate your analysis and results clearly.

The report should be concise. The marking of your report is based on the arguments presented and not the length. You may use bullet points to present your analysis.

3.4 Source Code

In addition to the written report, you are required to submit the source code that was used to generate your prediction results. The submitted code must be reproducible and well-documented.

Submission Requirements

- **Complete Source Code:** All scripts necessary to reproduce your results must be included in your submission.
- **README File:** Provide a **README.txt** that explains your program and any known problems.

Master script

You must include one of the following master files to reproduce your submitted prediction files from the released datasets:

- **Python:** include a file named **run.py**, executable using python **run.py** without errors.
- **Jupyter Notebook:** include a file named **run.ipynb**, executable from start to finish without errors.

The master script/notebook should reproduce Predict_1.xlsx and Predict_2.xlsx using your final selected preprocessing pipeline, model, and parameter settings. The label column in the test datasets must be excluded from predictor variables. It is not required to rerun all exploratory analyses, visualisations, or exhaustive hyperparameter searches.

Testing Before Submission

- Before submitting, you must verify that your program runs correctly.

4. Mark Allocation

The assignment is marked out of 100, distributed as follows:

Satisfactory Submission (10 marks)

Marks are awarded based on:

- All required files submitted correctly.
- Declaration form executed and submitted.
- Python code (or Jupyter Notebook) must run without errors and reproduce the submitted predictions.
- Overall presentation quality of the report.

Data Preparation (30 marks)

Assessed on how well you identify, justify, and address issues in the data. Examples include:

- Removal of irrelevant attributes,
- Handling duplicates,
- Dealing with missing entries,
- Correct treatment of data types,
- Scaling/standardisation,
- Addressing class imbalance,
- Any innovative preprocessing steps taken.

Data Classification (30 marks)

Assessed on the quality and clarity of:

- Model selection and justification,
- Training, tuning, and validation processes,
- Comparative evaluation of classifiers,
- Logical reasoning behind the final chosen models and predictions.

Prediction (30 marks)

Assessment is based on the average classification accuracy of the **final selected model** when applied to the two test datasets: Test-data-1.xlsx and Test-data-2.xlsx. In other words, the final score will reflect the accuracy of the predictions submitted in Predict_1.xlsx and Predict_2.xlsx.

Accuracy < 70% → 5 marks

70% ≤ Accuracy < 80% → 10 marks

80% ≤ Accuracy < 82% → 20 marks

82% ≤ Accuracy < 85% → 25 marks

Accuracy ≥ 85% → 30 marks

The final score is based on its average accuracy across the two test datasets. The formula is:

$$\text{Accuracy} = \frac{\text{Number of correctly classified instances}}{\text{Total number of instances in the test set}}$$

You will compute this separately on both test datasets, then take the average:

$$\text{Final Accuracy} = \frac{\text{AccuracyTest1} + \text{AccuracyTest2}}{2}$$

5. Submission

The assignment must be submitted in two parts as described below:

Part A: Main Report

- Submit a PDF version of your report via Turnitin (link available on Blackboard).
- File naming convention: report_surname_studentID.pdf
- On the cover page of your report, state the average classification accuracy obtained by your best-performing model across the two test datasets.

Part B: Supporting Files

- Submit all remaining materials as a single compressed .zip file via the assignment submission link on Blackboard.
- ZIP archive naming convention: surname_studentID.zip
- The .zip file must contain the following:
 - report_surname_studentID.pdf (same as Part A)
 - Signed declaration form (PDF)
 - Two prediction files:
 - Predict_1.xlsx (corresponding to Test-data-1.xlsx)
 - Predict_2.xlsx (corresponding to Test-data-2.xlsx)
 - Executable code:
 - Either run.py or run.ipynb that can reproduce Predict_1.xlsx and Predict_2.xlsx from the released datasets without errors, without using the test label column as a predictor.
 - A README.txt that explains your program (in case we cannot run your code).
 - Any other relevant supporting materials:
 - Model files (if applicable)
 - Visualizations (plots, screenshots, etc.)
 - Additional documentation or helper scripts

Important: Ensure that all files are placed directly in the root of the ZIP archive. Do not use subfolders. The ZIP archive must be self-contained and executable, allowing reviewers to reproduce your results.

6. Academic Misconduct Plagiarism and Collusion

Please note the following:

Copying material (from other students, websites or other sources) and presenting it as your own work is plagiarism. Even with your own (possibly extensive) modifications, it is still plagiarism.

Exchanging assignment solutions, or parts thereof, with other students is collusion. Engaging in such activities may lead to a grade of ANN (Result Annulled Due to Academic Misconduct) being awarded for the unit, or other penalties. Serious or repeated offences may result in termination or expulsion. You are expected to understand this at all times, across all your university studies, with or without warnings like this.

END OF ASSIGNMENT